

Higher education, gender, and job opportunities of engineering graduates in Ethiopia: An exploratory study

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ABSTRACT

This study investigates the relationship between higher education and entry into the labor market in Ethiopia, with a focus on gender and job opportunities for engineering graduates. It uses a tracer study design to collect and analyze quantitative and qualitative data from 158 engineering graduates, 18 employers, and 16 higher education teachers. The findings suggest that higher education study conditions, curriculum relevance, and acquisition of employment skills influenced opportunities for engineering graduates to find employment shortly after graduation. In contrast to much of the existing research, our small sample of 31 female engineering graduates were not disadvantaged in comparison to males when they moved into employment immediately following graduation. Recommendations are suggested for improving the quality and relevance of engineering education to address gender gaps and challenges in higher education and employment. Future research to substantiate the findings in this exploratory study is also needed.

1. Introduction

Higher education institutions have a key role to play in enhancing the employability of their graduates, especially in developing countries like Ethiopia, where there is a need for skilled and qualified human resources for economic development and social transformation (World Bank, 2019). Engineering graduates, in particular, are expected to have the skills and competencies that are relevant and responsive to the needs and demands of employers and society (Sharma, 2015). However, there may be various factors that affect the employability of engineering graduates, such as the quality of learning environments, the alignment of curricula, and the development of skills and competencies (Tomlinson and Holmes, 2017).

This study is informed by the global agenda of gender equality and women's empowerment, which is not only a human right but also a smart economic strategy (OECD, 2017). According to the World Bank (2019), closing gender gaps in education and employment can boost economic growth and reduce poverty. However, despite the progress made in increasing women's access to higher education and employment, there are persistent challenges and barriers that limit their full participation and potential in these domains (Karnes, 2019). According to the International Labour Organization (ILO, 2020), there is a global

gender gap in labor force participation rates of 27 %, with only 48 % of working-age women participating in the labor market compared to 75 % of men. Moreover, there are gender disparities in the fields of study and occupational choices of women, especially in high-demand and high-paying sectors of science, technology, engineering, and mathematics (STEM) (Charles & Thebaud, 2018). Just 34 % of women work in STEM disciplines (science, technology, engineering, and math) in the World, and men predominate in STEM majors in college. In some of the future's highest-paying and fastest-growing professions, such as computer science and engineering, there are notably large gender disparities (American Association of University Women, 2015). According to the 17th edition of the World Economic Forum (WEF) Global Gender Gap Index (2023), which benchmarks gender parity across 146 countries cross-analysis, it will take 162 years to close the gender gap in political empowerment, 169 years to close the gender gap in economic participation and opportunity, and 16 years to close the gender gap in educational attainment at the current rate of progress over the 2006–2023 span. The time to close the gender gap in health and survival is still unknown.

In Sub Saharan Africa, significant gender gaps still exist in many areas, including school enrollment (Lelei and Weidman, 2012), employment patterns (Shafiq et al., 2018), and labor force participation

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(Sabir, 2015), especially in business (Tamrat, 2020). While Ethiopia's higher education system has grown from two institutions in the 1990s to 44 in 2016, and total enrolment in undergraduate degree programs has increased from 420,387 in 2009–778,766 in 2016 (FDRE/Ministry of Education, 2017), the graduate unemployment rate has also increased from 17.50 % in 2012 to 19.1 % in 2018 (MOE 2012, 2018). A tracer study of 400 out of 1600 graduates carried out in 2017 by Bahir Dar Polytechnic College showed that women are more vulnerable to unemployment problems than men, with 55.3 % of women and 44.7 % of men among these graduates reporting having been unemployed for more than a year (Gizachew et al., 2022). Policymakers face a significant challenge as a result of the mismatch between labor supply and demand in Africa's formal sector, given that the continent's youth population is expected to double to over 830 million by 2050 (AfDB, 2016; Ayele et al., 2017). More specifically, in sub-Saharan Africa (SSA), youth unemployment is three times higher than adult unemployment (AfDB, 2016). According to Mekonnen Yimer et al. (2024), time series forecasting indicates that Ethiopia's higher education graduate unemployment rates will remain unchanged (.02) between 2019 and 2028. The gender gap persists in both access to higher education and labor market success. This is particularly evident in the field of engineering, which is required for the economic development and industrialization of Ethiopia (Yizengaw, 2018).

National policy in Ethiopia stresses that human resources development is to be strengthened by training competent and innovative people with special attention to engineering, technology and natural sciences (FDRE-ESDP IV, 2010; FDRE-ESDP V, 2015). This is to be accomplished through introducing high quality science and mathematics curricula at primary and secondary schools (Tiruneh et al., 2021) as well as the recently adopted policy of the 70:30 university intake ratio in favor of engineering (FDRE-GTP I, 2010). The rationale behind this emphasis on engineering is many-sided. For example, supranational institutions (e.g., the World Bank, IMF, UNESCO, etc.) stress the significance of STEM (science, technology, engineering, and mathematics) education and the need to meet higher education benchmarks (Woldegiyorgis, 2014) that include STEM enrollment targets in policy decisions (Molla, 2019) and address perceived skills gaps and labor market demands. Increased participation in engineering programs, according to policy makers, will aid in closing these gaps (World Bank, 2010).

The enrollment and graduation trends in engineering programs show a low and uneven participation of female students (19.4 % in 2016), and high dropout (45.2 % in 2016) and unemployment rates (43.6 % within one year of graduation). Closing gender gaps can also benefit macro-economic and financial stability (Nekatiheb, 2004) that promote higher and more sustainable economic growth (Weidman, et al., 2015, Section VIII), as well as economic diversification (Sabri, 2015). Hence, it is critical to study the extent to which factors associated with higher education influence job opportunities for engineering graduates for both theoretical and policy reasons (Weidman and Jacob, 2011; Jacob and Holsinger, 2008).

Despite the importance of the higher education and labor market nexus in Ethiopia, there is a lack of comprehensive and empirical studies that examine the graduate employability situation, especially for women engineers. Most of the existing studies do not provide a clear picture of the factors that affect the transition and integration of women engineering graduates into the labor market, e.g., the mismatch between skills learned in higher education and employers' expectations (Yizengaw, 2018), gender stereotypes and discrimination, as well as personal and institutional barriers. Moreover, there is a gap in the literature on the policies, strategies, and actions that enhance the employability and empowerment of women engineering graduates in Ethiopia, and how effective they are in addressing the challenges and opportunities that women engineers face in their career paths.

This study aims to fill these gaps through a quantitative analysis, supported by qualitative observations, of the higher education and labor market nexus in Ethiopia, with a special focus on gender differences in

job opportunities for engineering graduates. The study also intends to provide policy and practice recommendations for improving the quality and relevance of higher education as well as addressing the gender gaps and challenges in higher education and employment. The following research questions are addressed:

- 1) How does the instructional environment of higher education institutions (e.g., study conditions, facilities, curriculum relevance, and competencies or acquisition of employability skills) influence job opportunities of engineering graduates?
- 2) Is there a gender difference in how higher education affects the employability (job opportunity) of engineering graduates in the labor market over the period immediately after graduation?

2. Higher education and job opportunities for graduates

In the knowledge economy, rapid technological changes and changing skill requirements in the labor market influence changes in the higher education landscape (Altbach, 2005). Essentially, graduates must have strong intercultural competencies and a broad understanding of the world in addition to the more conventional discipline information they learned as undergraduates (Salmi, 2009). In this situation, the infrastructure of Science and Technology Institutions (STIs), both hard and soft, becomes essential to bridging graduates' knowledge and financial gaps (Teferre, 2006). Soft infrastructure refers to networks and information technology that facilitate access to management, intellectual and R&D resources, as well as critical data on knowledge sources, market opportunities, and possible partners across national borders. Examples of these networks and technologies include peer-learning, off-class contacts, and field-based engagement (Varghese, 2011). Hard infrastructure comprises, among other common facilities, libraries, workshops, science and technology parks, well-equipped centers, technology and business incubators, smart and furnished classrooms, etc. (Varghese, 2011). An engineering student's employment success in the labor market is influenced by their ability to study in favorable circumstances.

However, in Ethiopia there are problems identified with the out-of-date curriculum and instructional methods (Hunde, 2023; Yizengaw, 2004), a shortage of qualified teachers (Hunde, 2023), and inadequate infrastructure of higher education institutions (Ayalew et al., 2009; Hunde, 2023; Teferre, 2006), especially in engineering and technology programs. These issues and challenges are exacerbated by the under-developed relationships between higher education and the workforce (Hunde, 2023), institutions' inadequate capacity for research, and the tenuous research-industry linkages gap (Negash, 2006). Taken together, these operate as obstacles blocking the ability of universities to enhance the capability of their graduates to find appropriate employment (Hunde, 2023).

Higher education curriculum needs to be responsive to the interests of employers to equip graduates with required and up-to-date employability skills (Hunde et al., 202 & Altbach, 2005). Curricula offered in HEIs should be well designed with objectives that address both local and international demands (Montgomery, 2013 & World Bank, 2010). Objectives are glasses through which we view our desired path or final product (Asgedom, 2000). Educational objectives become the criteria by which materials are designed, contents are outlined, instructional procedures are developed, and tests and examinations are prepared (Asgedom, 2000).

Higher education programs can play a crucial role in preparing graduates for future success in the dynamic labor market by equipping them with essential competencies or skills of 21st C (Hunde, 2023). Competencies or employability skills include mastery of a discipline (Peeters et al., 2019), the ability to work across disciplines (National Academies of Sciences, Engineering, and Medicine, 2020), the ability to learn new knowledge quickly (Tomlinson and Holmes, 2017), teamwork skills (Louise, 2010 and Tiruneh et al., 2018), IT skills (Tamrat, 2020), interpersonal and communication skills (Tamrat, 2020), critical

thinking (Louise, 2010), and creative thinking and visioning skills of graduates (Peeters et al., 2019). In Ethiopia, however, there is a persistent misalignment between skills obtained by engineering graduates and those desired by employers (Teshome and Oumer, 2024; Yizengaw, 2018).

How can we encourage and support women to enroll in programs that can lower unemployment and broaden women's participation in the labor force? Traditionally, female students have favored the social sciences and humanities, while shying away from fields such as engineering and science, which offer more job opportunities (National Academies of Sciences, Engineering, and Medicine, 2020). Feminist theory suggests that higher education institutions could organize special sessions (Louise, 2010) for female students to increase their knowledge about the gender aspects of job creation and placement and how to overcome them, as well as raising their awareness about the issues of the job market (Tamrat, 2020). Despite government initiatives to increase female enrollment in higher education, males continue to dominate (Semela, et al., 2019).

Gender disparities are notable in numerous domains (Jacob & Cheng, 2005; Ilie and Rose) of employment and labor market participation. The labor force participation rate is a powerful tool for formulating employment, training, and human resource development policies (Sabir, 2015). In Ethiopia, persistent policy directions outlaw any form of gender-based discrimination and guarantee equal rights and obligations for both sexes (Anwar et al., 2012). According to the World Bank (2018), Ethiopia's overall urban unemployment rate is 19.1 %, with women experiencing a rate of 26.4 percent, more than double than that of men (12.2 percent). In a similar vein, the unemployment rate among young women and men is 30.9 % and 19.0 %, respectively (Tamrat, 2020). Another study on youth self-employment in Ethiopia showed the same pattern, with the unemployment rate among women in urban areas (28.1 %) almost twice that of men (14.4 %). In relation to this, DEHEMS (2012) argues that in the study of unemployment it is important to take into account the actual time spent in the current state. That is, longer unemployment may lead to higher human capital depreciation, but also the intensification of search if unemployment benefits diminish over time. The issue of time spent looking for work is also a concern. Graduates from metropolitan universities are more likely to obtain jobs in a short period of time following graduation than graduates from rural (periphery) colleges. This is especially true for women (Shrum and Wyer, 2012; and Shrum, et al., 2015).

Gender-based discrimination as well as other obstacles hinder women's labor market participation (Tamrat, 2020). Women, for example, struggle to find relevant training and education possibilities (Heyneman, 2003), as well as information and market channels (IFC, 2019). Government support in these areas should be consistent, long-term, and oriented to expanding women's employment and investment. Human capital theorists also suggest that higher education institutions should play a role (Magali, 2004). Addressing gender equity in higher education institutions (Porter, 2011; Acker, 1994) requires challenging and transforming those structures, cultures, practices, and policies that stand in the way. Given these disparities, the present research examines how content and delivery of instruction in higher education (e.g. curriculum, facilities and organization of teaching and learning, etc.) is related to job competencies developed by graduates that, in turn, affect their transition into the labor market. The research focuses on a sector of the labor market, engineering, that is particularly important for the development of Ethiopia (Yizengaw, 2004). It also addresses gender differences.

3. Conceptual framework for the study

The conceptual framework developed by Schomburg (2010, p. 12) was adapted for the present research because it was used to study transition of higher education graduates into the labor market in international perspective. Schomburg's framework posits that Input ("Study

conditions and provisions"), Process ("Teaching and learning"), and Output ("Knowledge, skills, motivation, grades") lead directly to career Outcomes ("Transition employment, work"). Weidman's (2006) socialization framework for understanding higher education outcomes has similar dimensions.

Fig. 1 shows the hypothesized conceptual framework used to examine the effects of engineering students' higher education experiences (Study Conditions, Curriculum Relevance) and outcomes (Employment Competences/Skills) on employment following graduation (Opportunities in Graduate Labour Market). The framework posits that Gender moderates the relationships between higher education and employability outcomes in the engineering labor market.

HES (Higher Education Study Condition) refer to facilities and student experiences with instructors and peers that influence the teaching and learning process in academic environments (Schomburg, 2010; Weidman, 2006). HEC (Higher Education Curriculum Relevance) refers to a set of courses or a content domain offered to students as their learning plan (Pellegrino, 2006). C (Competencies of Employability Skills Acquisition) refer to a set of graduate attributes (hard and soft skills) that influence graduates' employability (Hillage and Pollard, 1998). JO (Job Opportunity in the Graduate Labour Market) refers to how long it took the engineering graduates responding to the survey to find employment after graduation.

4. Research design

The study used mixed research methods which blend quantitative and qualitative research techniques and approaches into a single study (Creswell, 2012). In terms of the time ordering, this study collected and analyzed the quantitative data followed by the qualitative data. The study followed a tracer study, namely, a graduate or alumni survey that traces the activities of the graduates or previous students of an educational institution (Latif and Bahroom, 2010; Schomburg, 2003). This study traced the engineering graduates of 2012 just two years after graduation. The two-year period was chosen to avoid retrospection bias (Latif and Bahroom, 2010). Also, if done immediately after graduation (an exit study), it would not cover the whole process of transition optimally because the work situation for a short time after graduating may be unstable. Further, it is common for graduates to require some months to find a job.

5. Population and sampling techniques

The study targeted engineering majors graduating in 2012 from two regional Universities in Ethiopia with similar characteristics, as well as their employers and lecturers. The major fields included Electrical, Mechanical, Civil, and Chemical Engineering. Using Taro's (2010) formula, the study sampled 195 graduates out of 380, ensuring a 5 % margin of error and a design effect of 2. The sample was distributed proportionally across the fields and randomly selected within each field, giving every graduate an equal chance of being chosen (Bryman, 2012). The return rate was 80 %. Table 1 shows the sample distribution by field.

6. Data collection tools

This study used questionnaires and semi-structured interviews to collect data. Closed-ended items were adapted from the UNITRACE Tracer Study Manual (Schomburg, 2003) and a study of Ethiopian engineering graduates' skill acquisition (Negeri and Oumar, 2023). Open-ended questions and interview schedules were designed to provide supplementary explanatory information to supplement the empirical analyses. They were either linked directly to the closed-ended items in the graduate survey or derived generally from the literature review. Amharic was used for both the translation of interview questions and the interview itself in order to facilitate communication.

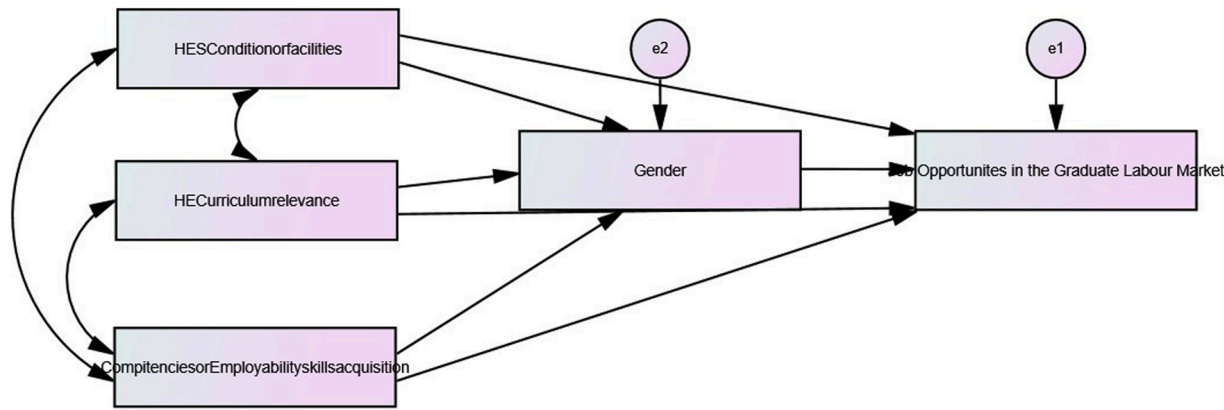


Fig. 1. Conceptual Framework for Higher Education Factors Associated with Job Opportunities of Engineering Graduates via Gender.

Table 1
Distribution of Graduate Samples in University 1 and University 2 in 2012.

Engineering Graduates in 2012 Fields of Study	University 1			University 2		
	M	F	Total	M	F	Total
Chemical Engineering	19	4	23	10	7	17
Electrical and Computer Engineering	15	5	20	20	6	26
Civil Engineering	20	6	26	38	8	46
Mechanical Engineering	18	4	22	12	3	15
Total	72	19	91	80	24	104
Grand Total 195						

Tables 2 and 3 show the closed-ended items used in the analysis. Questionnaires were administered through email, Facebook, mail, and hand delivery to 195 graduates of 2012 sampled from the Electrical, Mechanical, Civil, and Chemical Engineering fields at the two universities. Only 158 valid responses were obtained and analyzed. Interviews were conducted with 16 lecturers (two from each field and university) and 18 employers (six private, six public, and six self-employers) who were selected on the recommendations of the University-Industry linkage offices in both universities.

A pilot-test was carried out to evaluate the reliability and validity of the questionnaire and interview items (Creswell, 2012). The draft interview questions were given to three engineering lecturers and two economists. Based on their comments, several items were improved while others either stayed the same or were discarded. The estimated Cronbach Alpha for the Likert items was found to be 0.91, indicating high reliability.

Results from the confirmatory factor analysis (CFA) are shown in Fig. 2. Using at least 15 threshold values removed the variables of research skills (C6) and time management skills (C7); e14 and e15. All factor loadings are above 0.5 in the results depicted in Fig. 2. The model

also has unobserved variables that represent the errors of each observed variable. The figure displays the significant and positive factor loadings, the average variance extracted (AVE) values above 0.5, and correlation coefficients between 0.52 and 0.76. These values show that the indicators measure their latent variables well (HESC: Higher Education Study Conditions; HEC: Higher Education Curriculum; and C: Competencies perceived to have been acquired by graduates), and that the latent variables have convergent and discriminant validity. There were no statistical differences between institutions on item responses, either overall or by gender.

Calculated fit indices (Kline, 2015; Marsh & Hocevar, 1985) indicate that the model based on a maximum likelihood confirmatory factor analysis (CFA) of the covariance matrix for the three latent variables (HESC, HEC and C) is acceptable according to the Chi-square/degree of freedom ratio. Other acceptable fit indices for the model include an NFI (normed fit index) value of 0.895 (Bentler and Bonett, 1980); RFI (relative fit index) value 0.862 (Bollen, 1989); IFI (incremental fit index) value of 0.953 (Hu and Bentler); TLI (Tucker-Lewis index) of 0.963 (Hu and Bentler); CFI (comparative fit index) of 0.952 (Hu and Bentler); RMSEA (root mean square error of approximation) value of 0.068 (Browne and Cudeck, 1993); and Hoelter's critical N values (estimates how large the sample size should be for the Chi-square test) significant at or below the 0.05 level. which. In short, all goodness of fit parameters suggests that the data collected are robust.

To protect the identities of the participants, the universities involved in the study were given pseudonyms: University 1 and University 2. The university teachers who were interviewed had codes such as University 1 (Uni1Tech1, Uni1Tech2 to Uni1Tech6), and University 2 (Uni2Tech1 to Uni2Tech6). The employers who participated in the study were divided into three categories: Private Employers (PriEmp1 to PriEmp6), Public Employers (PubEmp1 to PubEmp6), and Self Employers (SlfEmp1 to SlfEmp6).

Structural equation modeling (SEM) was used to estimate the

Table 2
Survey Items Used to Construct Indicators of Higher Education Study Conditions (HESC) and Higher Education Curriculum (HEC).

To what extent did the Higher Education Study Conditions/facilities (HESC) in your higher education institution matter for your subsequent employment? 1 = very low; 2 = low; 3 = moderate; 4 = high; 5 = very high
HESC1 The facilities in the university (i.e., libraries, laboratories, workshops, etc.)
HESC2 Information and communication technology
HESC3 Off-class contacts with instructors (e.g., tutorials, mentoring services, projects, etc.)
HESC4 Contacts with fellow students (peer-learning)
HESC5 Engagement with field-based conferences, seminars, forums, workshops, etc.
How do you rate the relevance of the following curriculum components (HEC) in your higher education for remaining employable? 1 = very low; 2 = low; 3 = moderate; 4 = high; 5 = very high
HEC1 Objectives of courses/learning outcomes (i.e., if educational objectives are designed based on the national goals and socio-economic realities of the nation)
HEC2 Multi-disciplinarity is included in the content of courses to equip graduates with soft-skills and hard-skills from diverse areas.
HEC3 Methods of delivery (i.e., modern and active learning methods)
HEC4 Use of a variety of evaluation methods and assessment techniques
HEC5 Appropriate teaching materials to facilitate effective teaching-learning processes

Table 3
Survey Items Used for the Indicator of Competencies (C) Acquired by Graduates.

To what extent were the following competencies acquired while you were in higher education? 1 = very low; 2 = low; 3 = moderate; 4 = high; 5 = very high	
C1	Mastery of your own field of study or discipline
C2	Multidisciplinary knowledge competencies
C3	Ability to rapidly acquire new knowledge (i.e., alertness to new opportunities and know how to come up with new ideas and solutions)
C4	Teamwork skills (i.e., coordinating activities and managing tasks/negotiating skills)
C5	IT skills (computer skills)
C6	Research skills
C7	Time management skills
C8	Interpersonal and communication skills and emotional intelligence (including self-awareness, strength of character, confidence, motivation, leadership, and assertiveness)
C9	Critical thinking skills
C10	Creative thinking and visioning skills

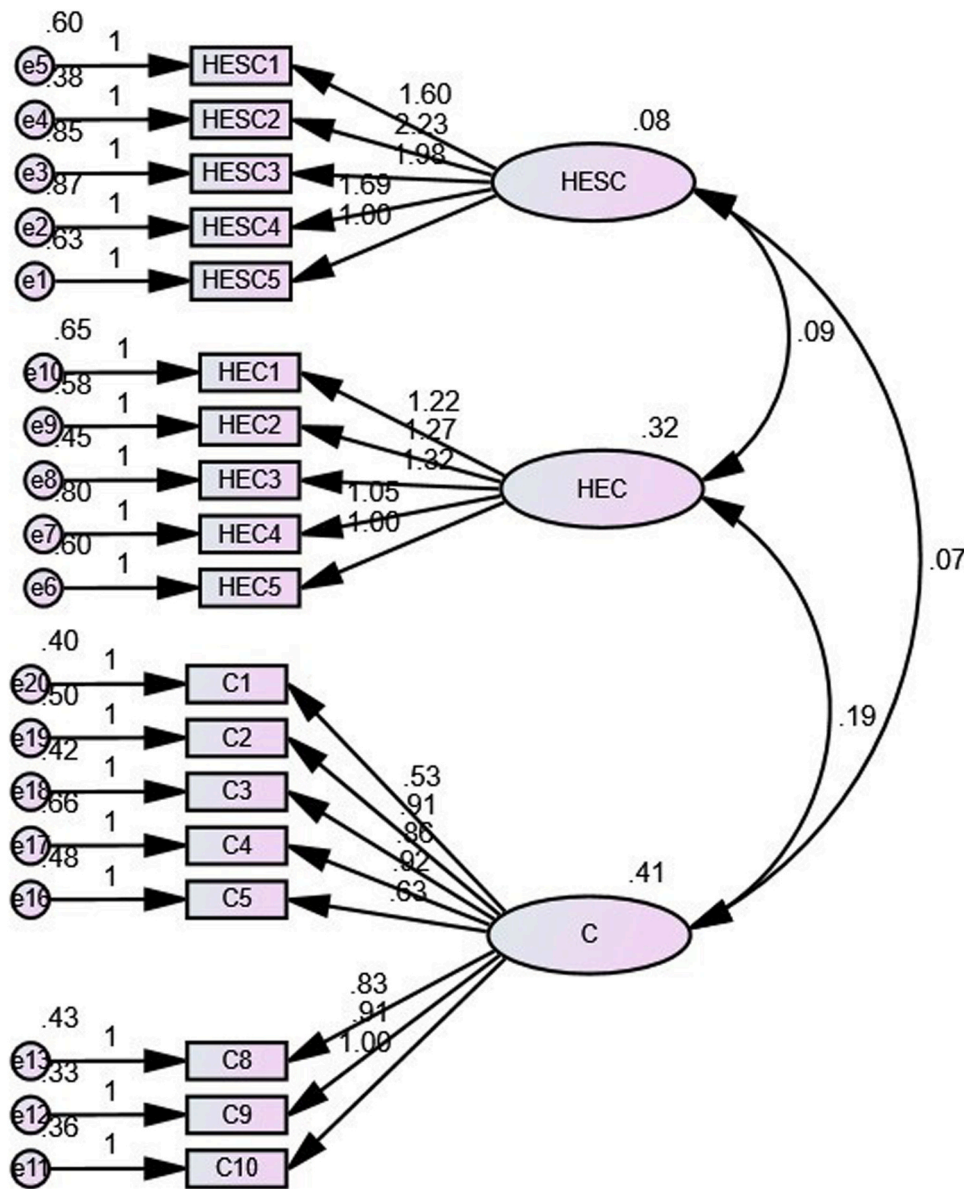


Fig. 2. Improved Confirmatory Factor Analysis (CFA) to Assess Reliability and Validity of Measures.

relationships among empirical measures through Amos and SPSS software. Qualitative data were organized through continuous comparison to identify common themes on each question (Creswell, 2012).

7. Findings

Reporting of results in this section is organized by research question.

Findings are discussed in the final section.

7.1. Effects of Ethiopian universities' instructional environments on the employment transitions of engineering graduates

Fig. 3 shows results from the structural equation model (SEM) based on the conceptual framework shown in Fig. 1 that posits the

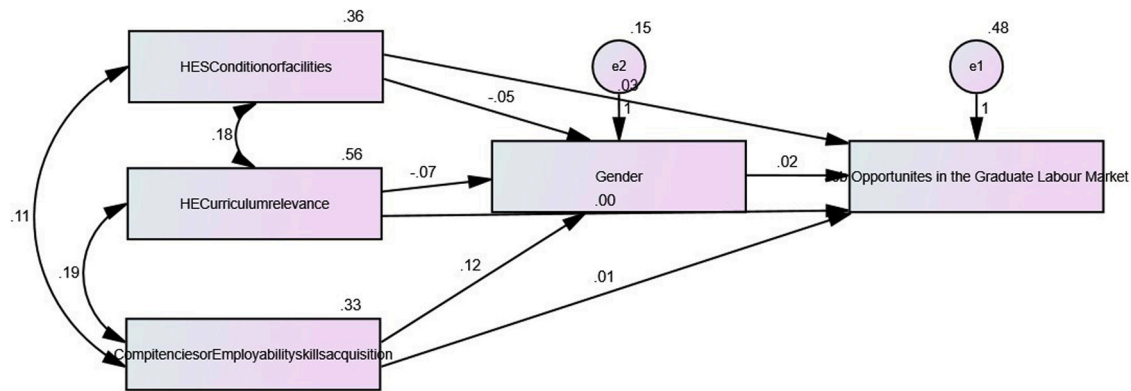


Fig. 3. Effects of Higher Education Study Conditions, Curriculum, and Competencies on Job Opportunities of Engineering Graduates via Gender.

relationships among higher education study condition (HESC), higher education curriculum relevance (HEC), competencies (C) or employability skills acquisition, and job opportunities (JO) of engineering graduates in Ethiopia. The model also tests how gender (G) mediates this relationship. Indicators showing how well the model fits the data include R-squared values of 0.36, 0.18, and 0.33 for higher education study condition (HESC), higher education curriculum relevance (HEC), and competencies (C), respectively. The error terms are 0.48 for e_1 and 0.15 for e_2 , which represent the unexplained variances in Job Opportunity (JO) and gender respectively. Job opportunity is an observed variable that measures how long it takes for graduates to find employment after graduation. It is divided into three categories: getting employment within six months (1), within a year (2), and more than a year (3). Thus, a higher score reflects taking a longer time to find employment immediately after graduation. Overall, findings suggest that the higher education variables in the model have very weak effects on how fast Ethiopian engineering majors can get jobs after graduation.

The only path that shows a statistically significant ($p < .01$) direct effect is higher education study conditions on job opportunities (0.037) after graduation. The positive effect indicates that graduates who rated their education experience most highly also took a longer time to find employment after graduation. This suggests a potential mismatch between what students valued most about their university preparation and what employers were seeking, a pattern found in other related research (Negeri and Oumar, 2023; Yizengaw, 2018). It may also indicate that Ethiopian engineering students were not very aware of what aspects of their experiences with the university instructional environment were most likely to help them find employment shortly after graduation.

To explore this further, university teachers were interviewed: "How well do Ethiopian universities support the study conditions of graduates that enable them to be competitive in the current job market?" The majority of university teachers (Uni1Tech2, Uni1Tech3, Uni1Tech4, Uni1Tech5, Uni1Tech6, Uni2Tech1, Uni2Tech2, Uni2Tech3, Uni2Tech4, and Uni2Tech6) agreed that the government is making efforts to expand and facilitate universities in every corner of the nation and they see this in their daily experiences. However, two university teachers (Uni1Tech1 and Uni2Tech5) had a different perspective. Specifically, Uni1Tech1 said: "I don't think so because the focus is not on the facilities but on the expansion of universities." Uni2Tech5 also expressed: "There are not enough laboratories, classes and dormitories, so I don't believe that the facilities are adequate and suitable for study." This suggests that university teachers think more investment is needed to equip universities with facilities that help graduates to prepare better for the future job market.

The direct effect of higher education curriculum relevance on job opportunities (0.009) was not statistically significant. Teachers were asked; in your opinion is the HE curriculum relevant for the current labor market? The majority of interviewed teachers (Uni1Tech1, Uni1-Tech2, Uni1Tech3, Uni2Tech3, Uni2Tech4, Uni2Tech5 and Uni2Tech6)

agreed that the curriculum in place has incorporated new demands on graduates towards becoming competitive in the labor market, in particular, and contributing to socio-economic development, in general. Respondents also said that the engineering curriculum has focused on skill-oriented and practice-based courses; however, some (Uni1Tech4, Uni1Tech5, Uni1Tech6, Uni2Tech1 and Uni2Tech2) proposed revisions to match the needs and demands of the current labor market. The teachers who were interviewed said they tried diligently to implement active learning methods and hand-on training in their classrooms, but they needed extra effort as well as additional instruction and better facilities to integrate theory and practice in their courses. Teachers recognized the importance of using approaches that provide graduates with the opportunity to actively participate in the learning process, apply theoretical knowledge, and develop practical skills that are applicable to real-world situations (Salmi, 2017).

The direct effect of competencies or employability skills on job opportunities (0.016) was also not statistically significant. In relation to this, teachers were asked: "Do you believe graduates acquire the employability skills (competencies) that the job market requires?" The majority of the teachers interviewed (Uni1Tech1, Uni1Tech5, Uni1Tech6, Uni2Tech1, Uni2Tech2, Uni2Tech3, Uni2Tech4, Uni2Tech5 and Uni2Tech6) agreed that graduates have the required engineering knowledge, wisdom, and skills necessary for successful practice as engineers, but that the nation needs a macroeconomic revolution to accommodate graduates on time and also to prevent unemployment and underemployment. They repeatedly mentioned that it's not about lacking competencies, but about the job market in a sluggish economy to absorb all the well-trained engineers and scientists. Only a few (Uni1Tech2, Uni1Tech3, and Uni1Tech4) had different opinions, focusing mainly on the necessity of providing distinct employability skill courses that equip graduates for the ever changing job market.

However, interviews with employers (PriEmp3, PriEmp4, PriEmp6, PubEmp5, SlfEmp2, SlfEmp3, SlfEmp4 and SlfEmp5) showed a big gap between employer expectations and graduate performance, particularly with respect to quality of work, productivity, and specific job-related knowledge and skills. When asked "Do you believe graduates have the employability skills (competencies) required by the labor market?" respondents from PriEmp5, PubEmp1, PubEmp3 and PubEmp2 said that fresh graduates were not competent enough to start their jobs without specific job-related training. Such training by companies ranged from a month up to six months' time. Of course, one of the benefits employer companies got from the graduates was the chance to have cheap labor. Respondents PubEmp6, SlfEmp1, and PriEmp1 and PriEmp2 suggested that some graduates lacked practical skills, appropriate attitudes for their jobs, and relevant knowledge of the job market. Respondents PubEmp4 and SlfEmp6 also remarked that many new graduates tended to lack professionalism, office discipline, integrity, willingness to learn new things, and reluctance to take responsibility for certain tasks. This implies that higher education institutions should be more receptive to

employer demands so that graduates can meet job requirements (Negeri and Oumar, 2023; Yizengaw, 2018).

7.2. Gender as an intervening variable between the instructional experience and employment prospects of engineering graduates

In Fig. 3, the indirect effects of higher education study condition (-0.05) and higher education curriculum relevance (-0.078) on job opportunities (JO) through gender (G) are negative, but not statistically significant. Because the gender variable assigns the higher value to males (2, as opposed to 1 for females), it is interesting to note that the negative coefficients imply that female engineering graduates may tend to view higher education study conditions and curriculum relevance in their universities more positively than their male counterparts.

However, the indirect effect of employability skills (0.124) on job opportunities (JO) through gender (G) is both positive and statistically significant at the 0.05-level. This finding suggests that males are more likely than females to have a positive view of the employability skills developed at their respective universities. When asked about the special support they provided for females in higher education as well as in the job market that helps with securing employment shortly after graduation, some interviewees (PubEmp1, PubEmp2, PubEmp3, PubEmp4, PubEmp5, PubEmp6, SlfEmp1, and SlfEmp2) shared common views. They had contributed, for example, by guiding females on their career options, introducing them to the job market through speeches (as alumni/ae and invited guests in universities) and contacts, aligning skills taught with the employers' expectations and demands, and providing female students with work-study opportunities in their companies to gain practical experience in their fields. But, the majority of respondents (PriEmp1, PriEmp2, PriEmp3, PriEmp4, PriEmp5, PriEmp6, SlfEmp3, SlfEmp4, SlfEmp5, and SlfEmp6, PriEmp4 and PriEmp5) said that no special support was provided for females.

The statistically significant ($p < .05$) coefficient of 0.026 indicates that gender affects job opportunities/transition to employment. It is interesting to note, however, that the positive coefficient suggests female graduates may tend to find employment somewhat faster than male graduates because gender (1 = female) and time to employment (1 = with 6 months) are both reverse coded. In fact, in these data, 25.8 % of females find employment within 6 months after graduation as compared to only 20.5 % if males. This difference, though favoring females, is not statistically significant. At worst, our data suggest that males have no advantage over females in their transition into employment after graduation. Consequently, our finding does not agree with other research (Negeri and Oumar, 2023; Yizengaw, 2018) indicating that Ethiopian female engineering graduates tend to be disadvantaged with respect to their male counterparts in finding employment.

In contrast, interviews with employers reinforced the belief that gender bias is a reality in the Ethiopian labor market. Most respondents (PriEmp1, PriEmp3, PriEmp4, PriEmp5, PriEmp6, PubEmp1, PubEmp2, PubEmp3, PubEmp4, PubEmp5, SlfEmp2, SlfEmp3, SlfEmp4, SlfEmp5, and SlfEmp6) admitted that there is likely to be some implicit or explicit gender bias in the job market that affects the job opportunities of female graduates, especially in male-dominated fields like engineering. They shared this view when interviewed about how gender is treated in the Ethiopian labor market. Only a few employers (PriEmp2, SlfEmp1, and PubEmp6) said that they do not discriminate or favor candidates based on their gender, but rather on their skills, qualifications, and competencies.

Interviewed teachers (Uni1Tech2, Uni1Tech3, Uni1Tech4, Uni1Tech5, Uni2Tech1, Uni2Tech2, Uni2Tech3, Uni2Tech5, and Uni2Tech6) also admitted that they did not offer any special support for women to pursue their academic and career goals. They said that they treated female graduates the same as male graduates, without considering their particular needs and challenges, but recognized that there were not any special policies or programs to promote gender equality and empowerment in higher education and the job market. Only a few university

teachers (Uni1Tech1, Uni2Tech4, and Uni1Tech6) said that they offered tutorials and entrepreneurship trainings for graduating students but admitted that it was not enough, and more work was needed in this area.

8. Discussion

8.1. Higher education and job opportunities of engineering graduates in Ethiopia

There were several findings in the present study that did not support expectations from the research literature. Specifically, of the three dimensions of the teaching/learning environment of the universities studied, only study conditions/facilities significantly affected early employment after graduation of Ethiopian engineering students. Ironically, more positive student assessments of study conditions/facilities had a negative effect on early employment after graduation. Interpreting this finding in terms of human capital theory, Becker (1993) suggests that investing in education can increase the skills, knowledge, and abilities of individuals (Salmi, 2017), which can enhance their productivity, employability, and income in the labor market (Becker, 2013; Wuttaphan, 2017). This finding also suggests that having universities catering more specifically to facility and curriculum needs of female engineering graduates may enhance their successful entry into the labor market after graduation.

Findings from the qualitative data show that, while most teachers acknowledged the government's efforts to expand universities, some teachers pointed out the lack of as well as shortcomings with laboratories, classrooms and dormitories. This suggests that more investment is needed in equipping Ethiopian universities with necessary facilities to help graduates prepare for the future job market. Based on our findings, we recommend for Ethiopia to invest more in higher education facilities so students can learn skills that last and matter (Semela, 2011). This study supports the OECD (2021) advice to make higher education inclusive, diverse, high-quality and relevant. Higher education should foster excellence and creativity for students to face the ever-changing labor market (Salmi, 2017; Weidman and Jacob, 2011).

The direct effect of the existing higher education curriculum relevance on job opportunities was not substantively significant. Most teachers think the existing curriculum is relevant, but some suggested revisions.

Empirical findings revealed that the direct effect of competencies or employability skills on job opportunities was not statistically significant, but interview data suggest that employability skills have notable effects on job opportunities. Most teachers think graduates have the skills, but some suggest more specific courses with required competencies (21st Century Skills). Some employers think graduates tend not to acquire competencies and need more training when entering the labor market. This implies that higher education institutions should adapt to employer demands in order to meet job requirements. This is supported by various sources (World Economic Forum, 2020; Chamorro-Premuzic and Frankiewicz, 2019) that emphasize the value of learning and intelligence for the future of work (Shafiq et al., 2018), with better higher education and employability skills leading to more job opportunities for graduates.

The foregoing results highlight the importance of taking a variety of individual and institutional characteristics into account when building programs in higher education. With respect to individual students, for instance, employability capital theory (Peeters et al., 2019) suggests that graduates can increase their chances of obtaining and maintaining employment by accumulating different forms of capital, such as human capital (skills, knowledge, abilities), social capital (networks, connections, relationships), cultural capital (values, norms, attitudes), and identity capital (self-concept, self-efficacy, self-esteem). Improving the relevance of higher education curriculum can increase graduates' employability capital by enhancing their human capital that is valued by employers; expanding their social capital that can provide them with

access and opportunities; developing their cultural capital can help them fit in and adapt to different contexts (Green et al., 2003), and strengthening their identity capital that can motivate them to pursue their goals and aspirations (Peeters et al., 2019; Bui and Nghia, 2022; Clarke, 2018). With respect to institutions, orienting at least some key aspects of academic programs toward preparation for employment as identified by potential employers could also be considered.

8.2. Gender and job opportunities of engineering graduates in Ethiopia

According to structural equation modeling (SEM), higher education variables have a positive and significant effect on job opportunities for engineering graduates, with gender partially mediating this effect. This suggests male and female graduates have different experiences and outcomes affecting their entry into the labor market. Despite the overarching belief that male graduates tend to find employment faster than female graduates, our data say otherwise. Nonetheless, in our interviews, most employers admit that there is gender bias in the job market, especially in engineering. However, a few employers say they do not discriminate by gender, but by skills, qualifications, and competencies. These findings are in line with global reports on gender disparity in engineering. The *Global Gender Gap Index*, (2021) by the World Economic Forum found that gender gaps are more likely in fields that require disruptive technical skills, such as engineering, where women make up only 20 % of the workforce. Moreover, women remain severely underrepresented in engineering, with just 12 % of those working in engineering being female.

The findings indicate that higher education factors such as study conditions, curriculum relevance, and competence acquisition are like bridges that connect graduates to job opportunities, while gender is like a gap that separates them. Interview data showed that there is limited support for women in higher education and a presence of gender bias and discrimination in the labor market. This finding is supported by Diehl et al. (2022) who found out that gender bias is still prevalent in gender-balanced and female-dominated industries, including higher education. More research is needed to explore more diverse, comprehensive, and relevant variables that can help understand and improve job opportunities for all engineering graduates in Ethiopia, not just females.

This study investigates how higher education shapes the future of engineering graduates in Ethiopia, with an emphasis on gender and job prospects immediately following graduation. We have identified factors that tell us part of the story, but they are not enough to comprehend the whole picture. Because there are other factors we have not considered that may have a bigger or different impact on job opportunities. Research is needed that can explore more diverse, comprehensive, and relevant variables that can help us understand and improve job opportunities for engineering graduates in Ethiopia. This includes exploring the role of other factors like job searching methods, personal characteristics, family background, labor market conditions, and social norms on job opportunities for Ethiopian engineering graduates.

Gender of Ethiopian engineering graduates is a key variable in this study. Surprisingly, our research suggests that female graduates were not disadvantaged with respect to their male counterparts in accessing and benefiting from higher education and employment opportunities. Nonetheless, both teachers and prospective employers believe that gender disparities persist. Of course, this is a very limited study with just 31 females in a study sample of 158. Even though this may not be a large enough sample to justify generalizing broadly from the results, the patterns identified do suggest in a very limited way that opportunities for female engineering graduates are improving. However, to continue moving forward, Ethiopia should invest more in improving the quality and relevance of higher education (Yizengaw, 2004; Ayalew, et al., 2009), addressing any gender gaps and challenges in higher education and the labor market, and fostering a culture of excellence and creativity among students and teachers. This suggests that policy and practice

should also tackle gender inequalities and prejudices persisting in both higher education and the Ethiopian labor market, as they are likely to influence the outcomes and experiences of male and female graduates differently.

Women in Ethiopia face many complex challenges (Tamrat, 2020; Semela, et al., 2019). Adding employment challenges can worsen the situation and limit possible benefits. However, there are various ways that higher education can help to bring change through closing gender gaps by enhancing and developing women's employability capital (World Bank, 2018). By facilitating study conditions in higher education (including facilities such as libraries, laboratories, and workshops), as well as the use of information and communication technology, off-class contacts, peer-learning, and engagement in field-based conferences, and seminars, women's human capital, social capital, cultural capital, and identity capital can be increased. According to Tamrat (2020), "higher education institutions could also prepare special sessions for female students to increase awareness about the gender dimensions of job creation and placement and how they could be overcome." Networking across higher education institutions can also facilitate understanding of global issues related to employment of women, including different cultures and values (Hays, 2008). Our research has shown that even though Ethiopian engineering professors are aware that female students should have more attention paid to their academic and employment situations, not much is happening to change this situation. While much remains to be done in Ethiopian higher education to overcome historical patterns of gender inequities, even among the most highly educated sectors of the labor market, ongoing research and policy initiatives are moving in promising directions (Negeri and Oumar, 2023; Teshome and Oumer, 2024).

Author statement

The corresponding author is responsible for the conceptualization, instrumentation, data collection, analysis, and revision of the work, while the co-author contributes to its design, writing, analysis, interpretation, rewriting, and language editing.

CRediT authorship contribution statement

Jerusalem Yibeltal Yizengaw: Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization. **John C. Weidman:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Formal analysis, Data curation.

Data availability statement

The data that support the findings of this study are openly available at <https://data.mendeley.com/datasets/4zsw4t7c3/1>

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